

```
docker run -d -p 8080:8080 --name airflow apache/airflow standalone
```

```
docker ps
```

```
docker logs airflow
```

```
docker exec -it airflow bash
```

```
cd /opt/airflow/dags
```

```
cat > student_marks_workflow_dag.py
```

```
from airflow import DAG
```

```
from airflow.operators.python import PythonOperator
```

```
from datetime import datetime
```

```
def create_marks_file():
```

```
    data = """Math,80
```

```
Science,75
```

```
English,90
```

```
Python,95
```

```
"""
```

```
    with open("/tmp/student_marks.txt", "w") as file:
```

```
        file.write(data)
```

```
    print("student_marks.txt file created")
```

```
def read_marks_file():
```

```
    with open("/tmp/student_marks.txt", "r") as file:
```

```
        lines = file.readlines()
```

```
        for line in lines:
```

```
            print(line.strip())
```

```

def calculate_total():
    total = 0
    with open("/tmp/student_marks.txt", "r") as file:
        lines = file.readlines()
        for line in lines:
            subject, marks = line.strip().split(",")
            total += int(marks)
    print(f"Total Marks = {total}")

def percentage_calculation():
    total = 0
    with open("/tmp/student_marks.txt", "r") as file:
        lines = file.readlines()
        for line in lines:
            subject, marks = line.strip().split(",")
            total += int(marks)
    percentage = total / 4
    print(f"Percentage = {percentage}")

def generate_result():
    total = 0
    with open("/tmp/student_marks.txt", "r") as file:
        lines = file.readlines()
        for line in lines:
            subject, marks = line.strip().split(",")
            total += int(marks)
    percentage = total / 4
    result = "PASS" if percentage >= 35 else "FAIL"
    with open("/tmp/result.txt", "w") as result_file:

```

```

    result_file.write("Student Result Summary\n")
    result_file.write(f"Total Marks = {total}\n")
    result_file.write(f"Percentage = {percentage}\n")
    result_file.write(f"Result = {result}")
    print("result.txt file generated")
with DAG(
    dag_id="student_marks_workflow_dag",
    start_date=datetime(2024, 1, 1),
    schedule=None,
    catchup=False
) as dag:
    create_marks_file_task = PythonOperator(
        task_id="create_marks_file",
        python_callable=create_marks_file
    )
    read_marks_file_task = PythonOperator(
        task_id="read_marks_file",
        python_callable=read_marks_file
    )
    calculate_total_task = PythonOperator(
        task_id="calculate_total",
        python_callable=calculate_total
    )
    percentage_calculation_task = PythonOperator(
        task_id="percentage_calculation",
        python_callable=percentage_calculation
    )

```

```
generate_result_task = PythonOperator(
    task_id="generate_result",
    python_callable=generate_result
)
```

```
create_marks_file_task >> read_marks_file_task >> calculate_total_task >>
percentage_calculation_task >> generate_result_task
```

```
airflow@0408891f5210:/opt/airflow$ cat /tmp/student_marks.txt
```

Math,80

Science,75

English,90

Python,95

```
airflow@0408891f5210:/opt/airflow$ cat /tmp/result.txt
```

Student Result Summary

Total Marks = 340

Percentage = 85.0

Result = PASSairflow@0408891f5210:/opt/airflow\$

exit

docker restart airflow

